

Artificial Intelligence - Weekly Assessment-1

1. What is a string? How can strings be created in Python?

A string in Python is an immutable sequence of characters used to store textual data.

Strings can be created using single quotes `'...'`, double quotes `"..."`, or triple quotes `'''...'''` or `"""..."""` (useful for multiline strings).

```
# Single quotes
s1 = 'Hello'

# Double quotes
s2 = "Python"

# Triple quotes (multiline)
s3 = """This is a
multiline string."""
```

2. What is string indexing? Explain positive and negative indexing.

String indexing allows you to access individual characters in a string using their positional index inside square brackets `[]`.

- **Positive Indexing:** Starts from `0` for the first character and moves left-to-right (`0, 1, 2, ...`).
- **Negative Indexing:** Starts from `-1` for the last character and moves right-to-left (`-1, -2, -3, ...`).

```
text = "PYTHON"

# Positive indexing
print(text[0]) # 'P'
print(text[2]) # 'T'
```

```
# Negative indexing
print(text[-1]) # 'N'
print(text[-2]) # 'O'
```

3. What is string slicing?

String slicing allows you to extract a substring from a string using the syntax `string[start:stop:step]`.

- `start`: Beginning index (included).
- `stop`: Ending index (excluded).
- `step`: Jump size (default is 1).

```
text = "Programming"

# Extract indices 0 to 6
print(text[0:7])    # "Program"

# Extract every second character
print(text[::2])    # "Pogamn"

# Reverse the string using negative step
print(text[::-1])   # "gnimmargorP"
```

4. Write a program to find the length of a user-entered sentence.

```
sentence = input("Enter a sentence: ")
length = len(sentence)

print("The length of the sentence is:", length)
```

5. Write a program to count the number of vowels in a string.

```
text = input("Enter a string: ")
vowels = "aeiouAEIOU"
count = 0
```

```
for char in text:
    if char in vowels:
        count += 1

print("Number of vowels:", count)
```

6. Write a program to reverse a string.

```
text = input("Enter a string: ")
reversed_text = text[::-1]

print("Reversed string:", reversed_text)
```

7. Write a program to check whether a word is a palindrome.

```
word = input("Enter a word: ").lower()

if word == word[::-1]:
    print("The word is a palindrome.")
else:
    print("The word is not a palindrome.")
```

8. Write a program to count how many times a particular character appears in a string.

```
text = input("Enter a string: ")
target_char = input("Enter the character to count: ")

count = text.count(target_char)

print(f"The character '{target_char}' appears {count} times.")
```

9. Explain the difference between upper, lower, title, and capitalize string operations.

These string methods transform character casing in different ways:

- **.upper()**: Converts all characters in the string to uppercase.
- **.lower()**: Converts all characters in the string to lowercase.
- **.title()**: Capitalizes the first letter of **every word** in the string.
- **.capitalize()**: Capitalizes only the **first letter of the entire string**, keeping the rest lowercase.

```
text = "hello python world"

print(text.upper())      # "HELLO PYTHON WORLD"
print(text.lower())      # "hello python world"
print(text.title())      # "Hello Python World"
print(text.capitalize()) # "Hello python world"
```

10. Write a program to remove extra spaces from the beginning and end of a sentence.

```
sentence = input("Enter a sentence with spaces: ")
cleaned_sentence = sentence.strip()

print("Cleaned sentence:", cleaned_sentence)
```